



Client Risks without Device

Understanding the Impact to and Continued Benefits
of Fraud Strategy After Gateway Partner Changes

JULY 13, 2020



Agenda



Overview

- Executive Summary
- State of Affairs
- Options to Consider

Understanding Impacts

- Overall Anonymous Client Risks
- Partner-Specific Risks

Continued Benefits

- Ongoing Value

Recommendations

Executive Summary



Introduction

IDVision with Digital unites Personal and Device Identities to get the Complete Picture of Client Credit Applicants



Situation

Gateway Partner wants to eliminate all JavaScript and to host the Client credit application page on their own server



Impact

IP Address and Device Identity info will no longer be available for Gateway Partner applications, affecting ability to link to historical identities



Assessment

Analyzed impact is low and our overall solution with recommended changes will continue to be effective in matching Gateway Partner applicants to existing identities

Gateway Partner to Fully Host Client Credit Application Page on internal Server without JavaScript

- No JavaScript means Device and IP would be missing from future Partner credit applications.

Device attributes provided by Gateway Partner are incomplete and not representative of the full swath of digital information captured



True geolocation of applicant's device provided by IP Address will be missing from risk assessment



With only the personal data of the applicant's PII available to assess risk, fraud management of Gateway Partner traffic is diminished



Strategy Under Consideration

- Integration Options for Mitigating Risks

1	Continue Current Integration Continue sending Gateway Partner traffic within the current integration point and configuration settings. The Gateway Partner traffic would not include device and IP.
2	Segment Gateway Partner Traffic Segment Gateway Partner traffic into a separate integration with modified configuration settings, facts, and weights to mitigate new fraud risks.
3	Gateway Partner Hosts 1st Party JS Continue efforts to pursue 1st-Party JavaScript Integration with Gateway Partner to create visibility into returning Gateway Partner devices within Gateway Partner traffic only.

Analyzing the Extent of Impact

Analysis Windows

In order to determine the impact of the Gateway Partner changes, our analysts assessed 2020 applications, knowing that Gateway Partner makes up 65% of Client overall traffic. There is no known segmentation of Gateway Partner traffic in our data.

For May, Client identified the Gateway Partner applications. This subset was used for side-by-side comparison.

Overall Client Traffic

JAN 2020 – MAY 2020

Gateway Partner vs Client Other

MAY 2020 Only

Continued Benefits
to Non-Partner
Traffic

Detect Known Fraud
with Reputation &
Histories

Realtime Analysis of
ID and Device Risk

Reduced Manual
Reviews

Impact of Gateway
Partner Changes

Geolocation

Identity Matching

Historical Insights

Manual Review

Gateway Partner Changes to Client Integration

Level-Set on True Impact



Impact of Gateway Partner Changes



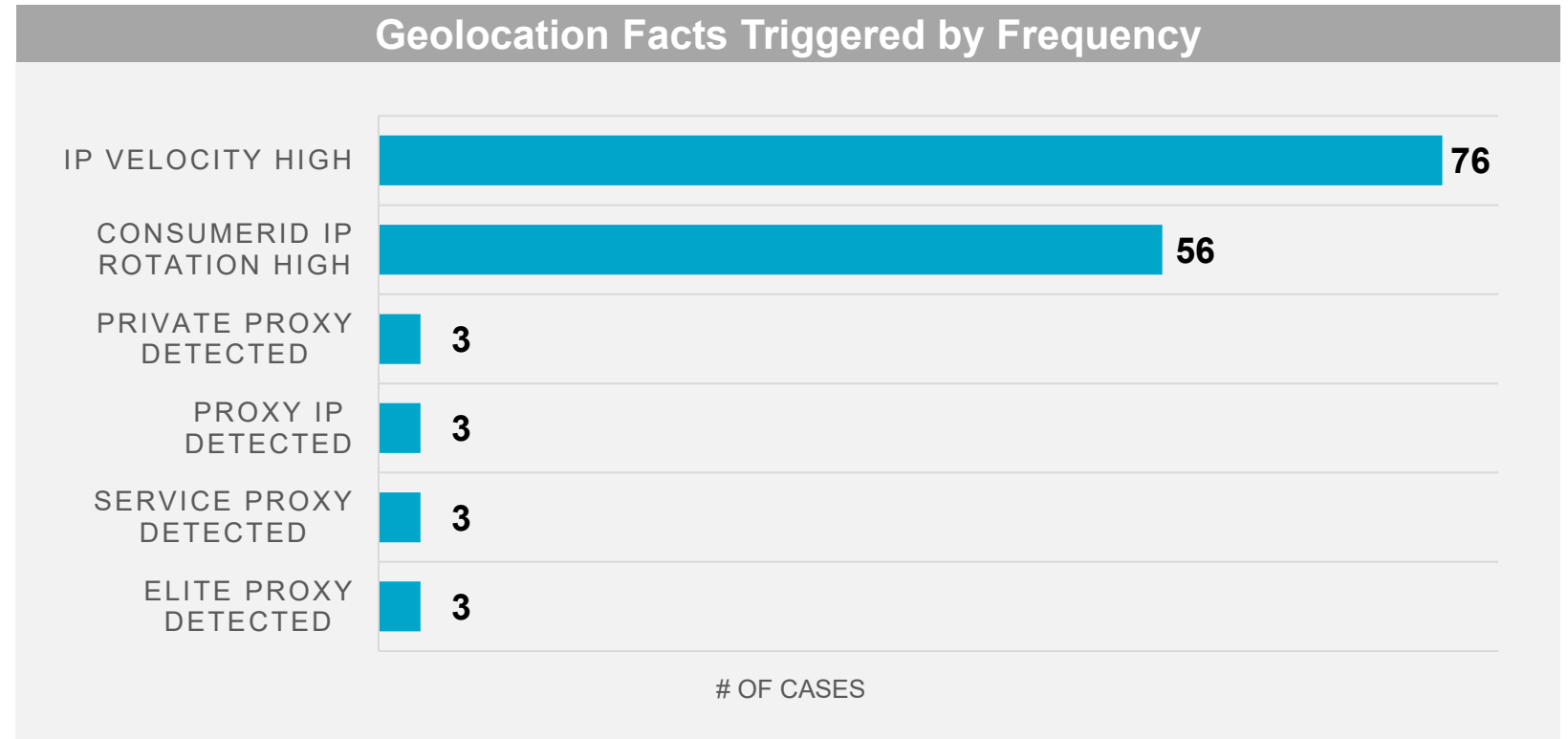
Increased risks from geolocation spoofing: Facts related to geolocation or IP attributes will no longer target risks within Gateway Partner traffic.

Geolocation Risks:

Geolocation-specific business restrictions cannot be enforced without IP to reveal the true location of the device.

Facts that evaluate IP Address, ISP, or Country for risk, velocity, or watch list comparisons will no longer be effective for Gateway Partner traffic.

Impact: **Low** ⬇️



* Analysis Window: JAN 2020 – MAY 2020



Impact of Gateway Partner Changes



Device & IP are used in the current configuration when matching an applicant to previously-known consumers. More applications will generate new ConsumerIDs.

Identity Risks:

Fraudsters tend to cycle through device attributes that will ensure they are new.

Fraudsters tend to check credit scores before application and submit multiple applications using different identities.

We cannot use device to link applicants that tumble attributes and identities.

Impact: **Low** ⬇️

	Gateway Partner Only	Client Other (Non-Gateway Partner)	Total
Total Cases	15,903	11,643	27,546
Total Cases Dependent Upon Device or IP	59	196	255
Device and IP Match	32	120	152
Device Match (no IP)	24	73	97
IP Match (no Device)	3	3	6
% of Cases Dependent Upon Device or IP	0.37%	1.68%	0.93%

Analysis window: MAY 2020



Impact of Gateway Partner Changes



We anticipate other additional impacts to operational costs and missed fraud-catch opportunities from the Gateway Partner change.



Reduced Historical Insights about Returning Devices

- Lack of common attributes between device and IP behavior within the platform for Gateway Partner devices first seen



Increased Fraud from Synthetic / Stolen Identities

- ID combined with Device provide powerful pillars of protection in global fraud management strategies
- Removing device from Gateway Partner traffic relies heavily on a single fraud solution



Possible increase in Velocity Facts Triggering

- Facts targeting velocity behavior could trigger more often
- TU will work closely with Client fraud team to establish correct facts and weights to maximize fraud catch without overburden to the review team

Impact: Unknown ➡

Gateway Partner Changes to Client Integration

Continued Benefits to Non-Gateway Partner Traffic

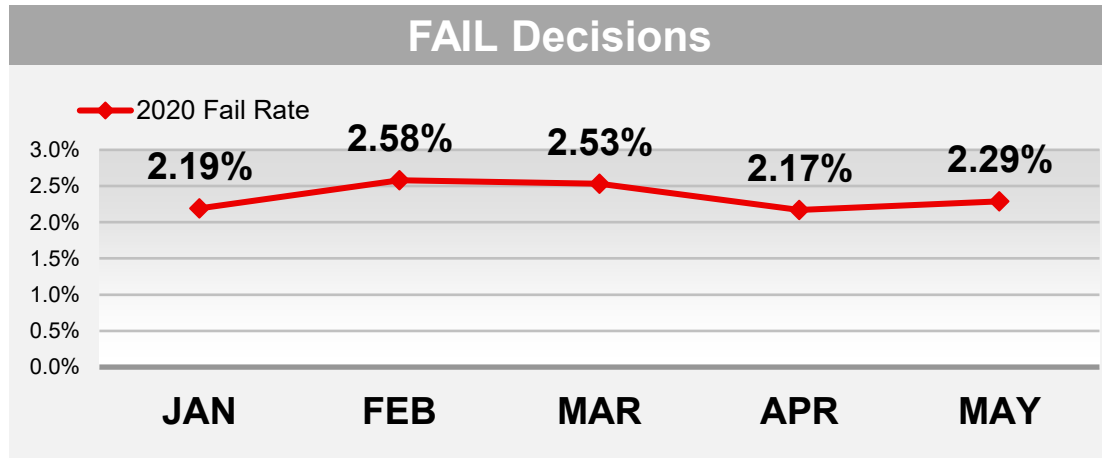


Continued Benefits to Non-Gateway Partner Traffic



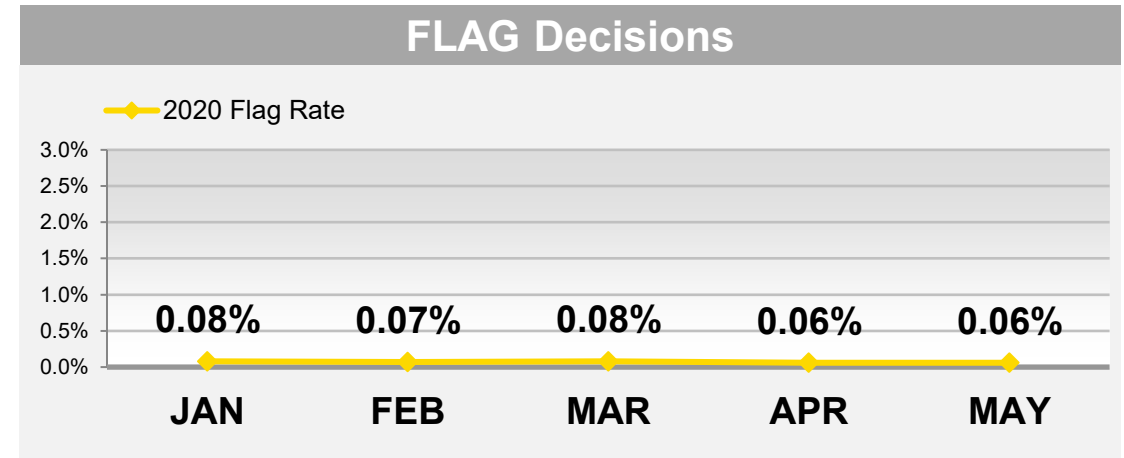
Identity with digital device fraud solutions in use across all other traffic and will have maximum fraud protection regardless of what integration options are chosen.

IDV stopped **3,506 applications** by bad actors and returning fraudsters in real time



* Analysis Window: JAN 2020 – MAY 2020

IDV expedited transaction processing by reducing manual review to **110 applications**



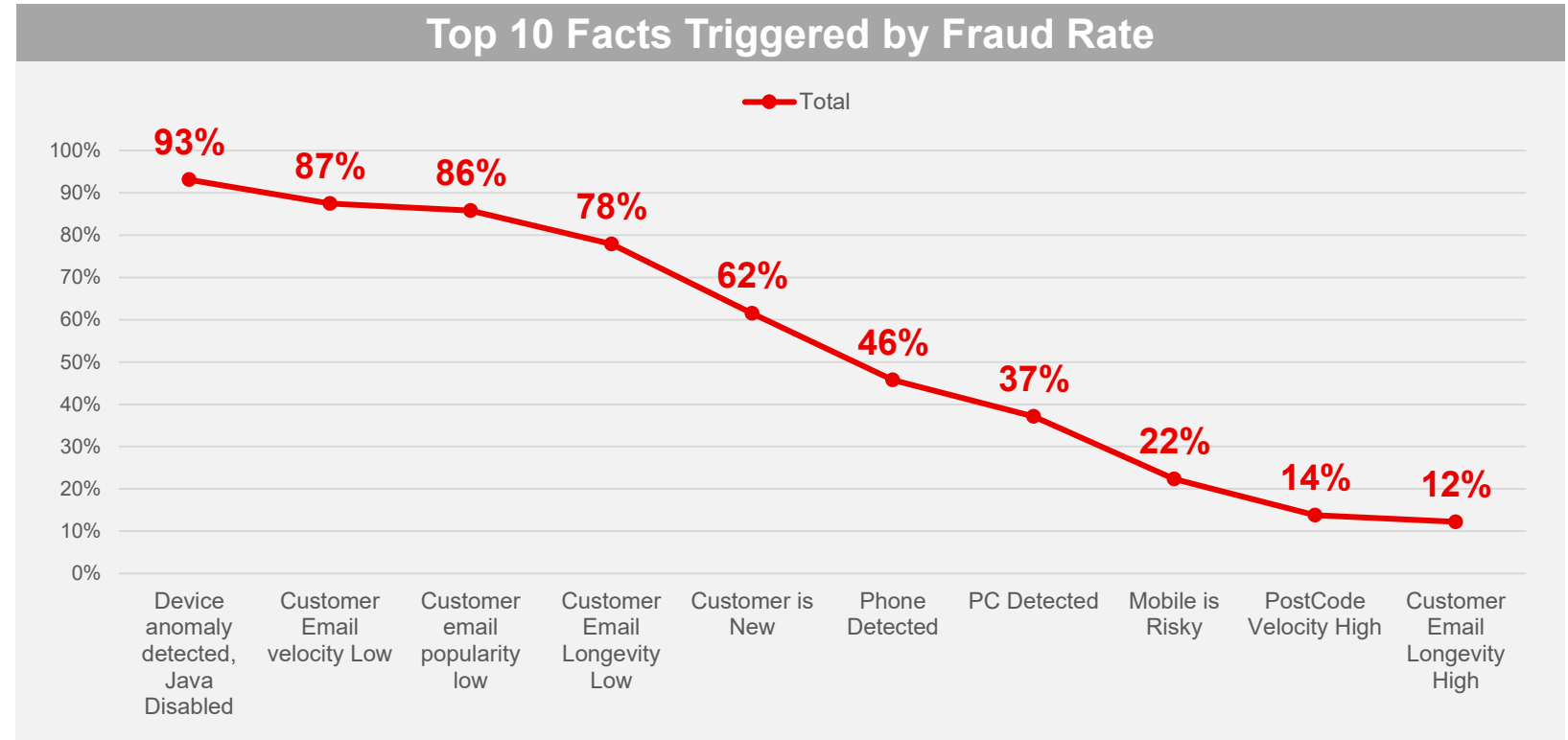
* Analysis Window: JAN 2020 – MAY 2020



Email Verification results continue to provide high value for all applications, and those facts have high correlation to confirmed fraud with low false positives.

Highly Effective Email Facts

- **Email Velocity Low**
 - 87% Fraud
- **Email Popularity Low**
 - 86% Fraud
- **Email Longevity Low (age)**
 - 78% Fraud



* Analysis Window: JAN 2020 – MAY 2020



Our fraud solutions help balance operational costs against fraud losses by returning real-time Decisions on new applications.

Summary of Decisions

We use a combination of facts to segment fraudsters from suspicious characters.

We continually tune your fact weights to address resource concerns and optimize fraud catch.

* Analysis Window: MAY 2020

	Gateway Partner	Client Other
Pass	98.2% 15,617 cases	96.9% 11,280 cases
Flag	0.03% 5 cases	0.1% 11 cases
Fail	1.77% 281 cases	3.0% 352 cases
Total Cases	15,903	11,643



Your feedback on Status and case management allows us to better tune our solution to target suspicious behaviors unique to your customer base.

Summary of Statuses

A hefty percentage of applications have been canceled before completion.

Remaining cases not shown have not be updated from the default status of *Place*.

* Analysis Window: MAY 2020

	Gateway Partner	Client Other
Good	23% 3,661 cases	31% 3,597 cases
Cancel	71% 11,338 cases	56% 6,545 cases
Fraud	0.23% 35 cases	0.44% 52 cases
Total Cases	15,903	11,643

Gateway Partner Changes to Client Integration

TU Recommendations



Send Gateway Partner application traffic to a new integration point so that we can modify the configuration in phases to rely less upon device & IP.

Phased Approach

As it seems unlikely that Gateway Partner will approve 1st party in the timeframe which is needed for this integration, we recommend a phased approach to integration changes.

- ☐ **Phase 1:** Send Gateway Partner application traffic to a new integration point with an alternate configuration, which relies less on device & IP for matching existing ConsumerIDs and weights more appropriately the risk for email we've noted
 - Increase email and phone matching weights in config
 - Remove *device attributes* and *device_id* from config
- ☐ **Phase 2:** Continue to work through Gateway Partner's approval process for 1st party JS
- ☐ **Phase 3:** Augment Gateway Partner traffic with device when it becomes available through the 1st party JS



Recommendations to Mitigate Concerns

Add IDVision with digital solution to Client LOGIN page so that we can check for device insights outside of account origination activity

Longer-term Suggestion: Enhanced protection and added visibility at login will continue to build on the pillars for fraud mitigation across all traffic and will provide even stronger protections to both Client and Gateway Partner subset of consumers.



Recommendations to Mitigate Concerns

Add KBA (Knowledge Based Authentication) capability through separate platform to further verify Gateway Partner identities at application



Recommendations to Mitigate Concerns



Increase the weight of the email verification low facts that improve fraud catch and change outcome for high-risk behavior.

Immediate Impact

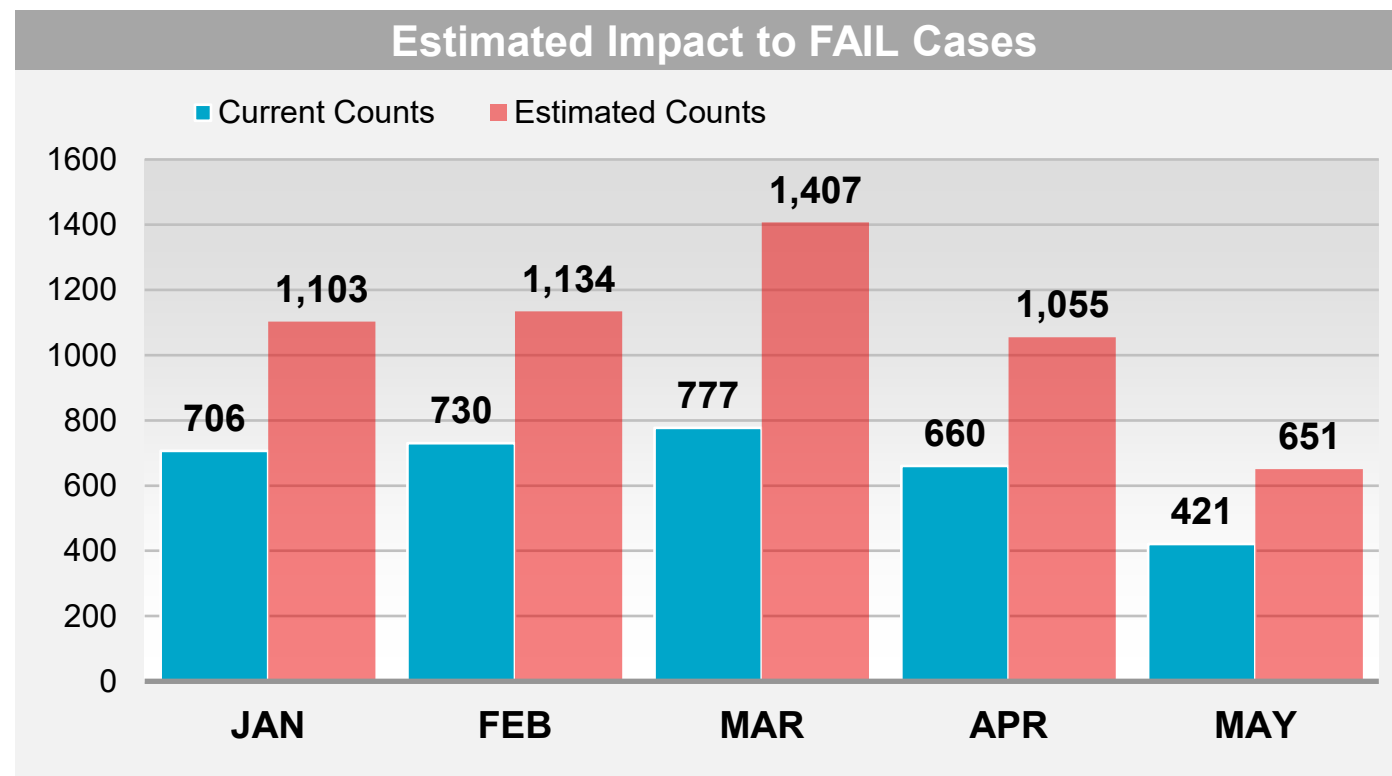
Our analysis shows that the **Email Age Low + Email Velocity Low + Email Popularity Low** combination is effective in identifying over 80% of highly-suspicious behavior in your customer base and will lessening your dependency on device+IP in Gateway Partner traffic.

Recommended Weight:

+ 124

Additional FAIL Cases:

+ 2,056 (increase to 3.59% fail rate)



* Analysis Window: JAN 2020 – MAY 2020



Recommendations to Mitigate Concerns



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Immediate Impact

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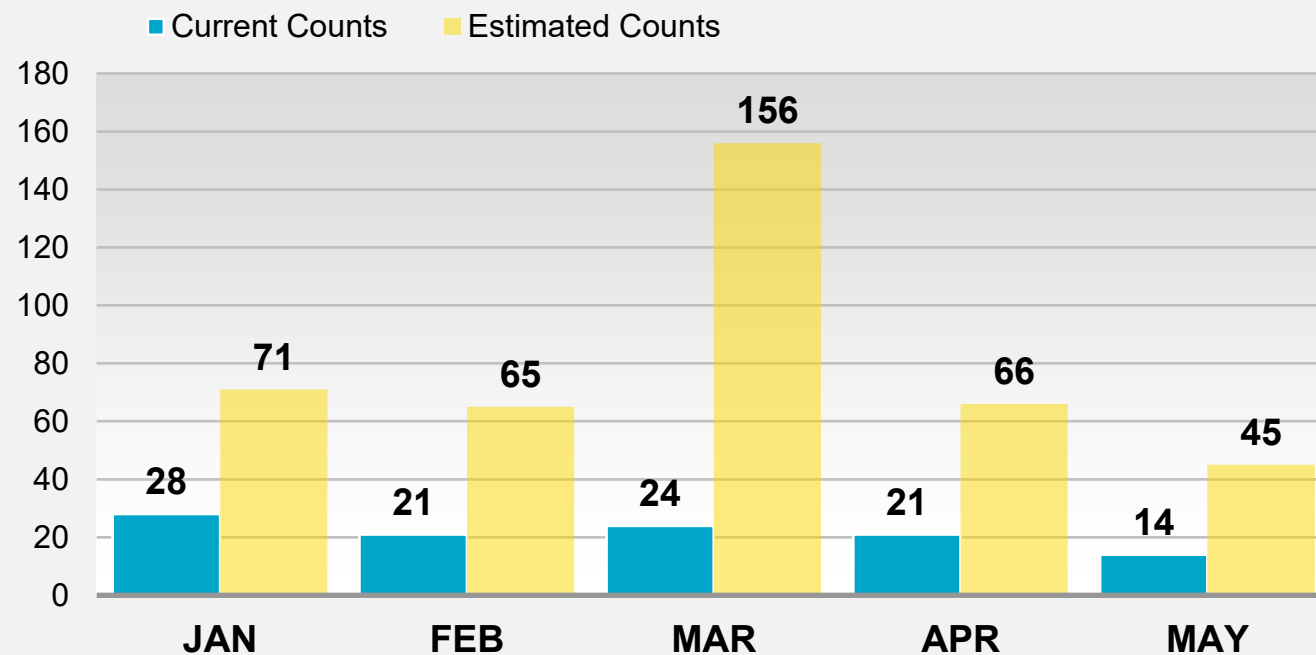
Recommended Weight:

+ 124

Additional FLAG Cases:

+ 295 (increase to 0.27% flag rate)

Estimated Impact to FLAG Cases



* Analysis Window: JAN 2020 – MAY 2020

Q&A